

Short-term effects of combinations of heart failure therapies on blood pressure, kidney function and serum potassium

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Among the most commonly cited reasons for failure to initiate comprehensive medical therapy in patients with heart failure are concerns relating to hypotension, kidney dysfunction and hyperkalemia. Here we performed a pooled individual participant-level analysis and developed a prediction model to estimate the short-term (2–12 weeks) treatment effects of combination medical therapy on systolic blood pressure (SBP), diastolic BP, estimated glomerular filtration rate (eGFR) and serum potassium. A total of 38,753 participants (16,877 with heart failure with reduced ejection fraction (HFrEF) and 21,876 with heart failure with mildly reduced or preserved ejection fraction (HFmrEF/HFpEF)) across nine randomized trials were included in the analysis, which tested angiotensin receptor blocker–neprilysin inhibitor (ARNI), steroidal mineralocorticoid receptor antagonist (sMRA), nonsteroidal MRA (nsMRA) and sodium glucose cotransporter-2 inhibitors (SGLT2i). For HFrEF, the estimated mean (95% prediction intervals (PIs)) treatment effect on SBP with combination ARNI + SGLT2i + sMRA therapy was -9.2 (-10.6 to -7.8) mmHg. For HFmrEF/HFpEF, the estimated mean (95% PI) treatment effects on SBP with SGLT2i + sMRA therapy and with SGLT2i + nsMRA therapy were -5.9 (-7.3 , -4.6) and -4.8 (-5.7 , -4.0) mmHg, respectively. For HFrEF, the estimated treatment effects of ARNI + SGLT2i + sMRA on eGFR and serum potassium were -6.8 (-7.9 , -5.4) $\text{ml min}^{-1} \text{m}^{-2}$ and $+0.30$ (0.25 , 0.35) mmol l^{-1} , respectively. For HFmrEF/HFpEF, the treatment effects for combination SGLT2i + sMRA and SGLT2i + nsMRA on eGFR were -7.7 (-8.9 , -6.4) and -6.1 (-6.7 , -5.5) $\text{ml min}^{-1} \text{m}^{-2}$, respectively and for serum potassium were $+0.34$ (0.29 , 0.38) and $+0.21$ (0.18 , 0.25) mmol l^{-1} , respectively. These analyses provide individualized estimates of the expected treatment effect for any combination of medical therapies in HFrEF and HFmrEF/HFpEF on BP, kidney function and serum potassium.

The prevalence of heart failure (HF) continues to rise and is associated with significant mortality and morbidity¹. Over the past three decades, there have been major advances in pharmacotherapy for patients with HF with reduced ejection fraction (HFrEF) and HF with mildly reduced or preserved ejection fraction (HFmrEF/HFpEF). For HFrEF, most patients are treated with a β -blocker, together with angiotensin-converting enzyme inhibitors (ACEI) or angiotensin receptor II blockers (ARB)^{2–4}. However, the use of more comprehensive medical therapies such as angiotensin receptor blocker–neprilysin inhibitors (ARNI), sodium glucose cotransporter-2 inhibitors (SGLT2i) and mineralocorticoid receptor antagonists (MRA) remains poor, and registry studies suggest <2% of eligible patients with HFrEF are on the recommended four-pillar medical therapy with β -blocker, ARNI, steroidal MRA (sMRA) and SGLT2i^{2,5}. The benefits of these pharmacological therapies appear to be additive, and combined use of four-pillar medical therapy is associated with a 60% reduction in all-cause mortality⁶. More recently, pharmacological therapy has become available for HFmrEF/HFpEF, with the advent of SGLT2i and the nonsteroidal MRA (nsMRA) finerenone. The use of combination therapy in HFmrEF/HFpEF is expected to significantly reduce the risk of cardiovascular death and worsening HF events, where substantial lifetime benefits can be achieved with sustained use⁷. As new treatments continue to be developed, there is an urgent need to bolster implementation efforts to improve the use of medical therapies across the spectrum of left ventricular ejection fraction (LVEF).

Current guidelines recommend titration of medical therapy as frequently as every 1–2 weeks depending on patient symptoms, vital signs and laboratory findings⁸. The safety, tolerability and efficacy of uptitration of guideline-directed medical therapies for acute heart failure (STRONG-HF) trial showed that rapid uptitration of medical therapy after hospitalization for acute HF reduced the incidence of death and HF hospitalization at 6 months compared to usual care⁹. Despite growing interest in rapid initiation of HF therapy, uptake remains poor¹⁰. Concerns of adverse events, particularly hypotension, kidney dysfunction and hyperkalemia, are the most commonly cited reasons for therapeutic inertia^{11,12}. Understanding the expected treatment effects of medical therapy on blood pressure (BP), kidney function and serum potassium levels may allow clinicians to more rapidly initiate and uptitrate combination medical therapy, either simultaneously or in rapid sequence.

Access to individual participant-level data allows for refined estimates of treatment effects according to common baseline characteristics, such as age, sex, body mass index (BMI) and kidney function, and BP. Therefore, we conducted an individual participant-level analysis of pivotal randomized double-blind controlled trials to estimate the treatment effects of comprehensive medical therapy (ARNI, SGLT2i, sMRA or nsMRA) in HFrEF and HFmrEF/HFpEF on (1) short-term changes in BP, estimated glomerular filtration rate (GFR) and serum potassium levels and (2) cardiovascular death and worsening HF events. We also aimed to develop a calculator that estimates the treatment effect of these clinical outcomes for combination medical therapy as a clinical tool to guide prescribing.

Results

The primary analysis included 38,753 participants (16,877 with HFrEF and 21,876 with HFmrEF/HFpEF) across the following nine randomized trials: DAPA-HF ($n = 4,744$), DELIVER ($n = 6,263$), PARADIGM-HF ($n = 8,442$), PARAGON-HF ($n = 4,822$), RALES ($n = 1,663$), TOPCAT-Americas ($n = 1,767$), FINEARTS-HF ($n = 6,001$), CHARM-Alternative ($n = 2,028$) and CHARM-Preserved ($n = 3,023$; Table 1). Of the primary trials evaluating ARNI, sMRA, nsMRA and SGLT2i, background use of β -blockers and ACEI/ARB was high at >70% (other than the RALES trial, which had low background β -blocker use). The treatment effect of ARNI, sMRA, nsMRA or SGLT2i on SBP, diastolic BP (DBP), eGFR or serum potassium were broadly consistent across subgroups defined by background medical therapy use (Supplementary Table 1).

In HFrEF, the estimated mean (95% prediction interval (PI)) treatment effect on SBP with the three-drug combination of switching to ARNI (from ACEI/ARB), starting an SGLT2i and starting sMRA was -9.2 (-10.6 , -7.8) mmHg. In HFmrEF/HFpEF, the estimated mean (95% PI) SBP effect with SGLT2i + sMRA and SGLT2i + nsMRA was -5.9 (-7.3 , -4.6) and -4.8 (-5.7 , -4.0) mmHg, respectively (Fig. 1). Among patients with HFmrEF/HFpEF and LVEF < 60%, the estimated mean (95% PI) SBP effect with ARNI (from ACEI/ARB) + SGLT2i + sMRA and ARNI (from ACEI/ARB) + SGLT2i + nsMRA was -10.3 (-11.8 , -8.9) and -9.3 (-10.3 , -8.3) mmHg, respectively. Extended Data Fig. 1 shows the estimated treatment effects of combination therapy on DBP.

In HFrEF, the mean (95% PI) treatment effect for three-drug combination of ARNI (switch from ACEI/ARB) + SGLT2i + sMRA in eGFR was -6.8 (-7.9 , -5.4) ml min⁻¹ 1.73 m⁻² (Fig. 2). In HFmrEF/HFpEF, the two-drug combinations of SGLT2i + sMRA and SGLT2i + nsMRA had a mean treatment on eGFR of -7.7 (-8.9 , -6.4) and -6.1 (-6.7 , -5.5) ml min⁻¹ 1.73 m⁻², respectively. In HFrEF, ARNI (from ACEI/ARB) + SGLT2i + sMRA was associated with a treatment effect on serum potassium of $+0.30$ (0.25, 0.35) mmol l⁻¹ (Fig. 3). The mean treatment effect on serum potassium for SGLT2i + sMRA and SGLT2i + nsMRA in HFmrEF/HFpEF was $+0.34$ (0.29, 0.38) and $+0.21$ (0.18, 0.25) mmol l⁻¹, respectively. Among patients with HFmrEF/HFpEF and LVEF < 60%, the three-drug combinations of ARNI (from ACEI/ARB) + SGLT2i + sMRA and ARNI (from ACEI/ARB) + SGLT2i + nsMRA were associated with mean treatment effects on serum potassium of $+0.33$ (0.28, 0.38) and $+0.20$ (0.16, 0.24) mmol l⁻¹, respectively.

Estimates for starting ARNI de novo rather than switching from existing ACEI/ARB therapy are presented in the Extended Data (Extended Data Figs. 2–5). In HFrEF, the estimated treatment effects with the three-drug combination of ARNI (de novo) + SGLT2i + sMRA were -14.9 (-16.6 , -13.1) mmHg for SBP, -8.1 (-9.2 , -6.9) mmHg for DBP, -9.5 (-11.5 , -7.4) ml min⁻¹ 1.73 m⁻² for eGFR and $+0.44$ (0.38, 0.51) mmol l⁻¹ for serum potassium. The estimated treatment effect of combination therapy on BP, kidney function and serum potassium stratified by sex is shown in Supplementary Figs. 1–6.

The validation analysis was derived from 1,016 participants from four randomized clinical trials (including two crossover trials) from INITIATE ($n = 62$), SOGALDI-PEF ($n = 108$), CONFIDENCE ($n = 800$) and ROTATE-3 ($n = 46$; Supplementary Table 2). The root mean square error for SBP, DBP, eGFR and serum potassium was 2.4 mmHg, 1.6 mmHg, 1.0 ml min⁻¹ 1.73 m⁻² and 0.1 mmol l⁻¹, respectively. The mean absolute error for SBP, DBP, eGFR and serum potassium was 1.9 mmHg, 1.4 mmHg, 0.8 ml min⁻¹ 1.73 m⁻² and 0.07 mmol l⁻¹, respectively. Visually, there was good correlation between observed and estimated treatment effects (slope of regression = 1.13, 95% CI = 0.99–1.28; Supplementary Fig. 7).

The treatment effects of combination therapy on cardiovascular death or first worsening HF events are shown in Fig. 4. In HFrEF, three-drug combination ARNI (switching from ACEI/ARB) + SGLT2i + sMRA is associated with a 61% (52%, 68%) reduction in the composite endpoint compared to standard treatment. In HFmrEF/HFpEF, combination SGLT2i + sMRA and SGLT2i + nsMRA are both associated with 31% reductions in the composite endpoint. Finally, in the subgroup of HFmrEF/HFpEF with LVEF < 60%, three-drug combination ARNI (switching from ACEI/ARB) + SGLT2i + sMRA and ARNI (switching from ACEI/ARB) + SGLT2i + nsMRA are both associated with a 40% reduction in the composite endpoint. Figure 5 compares the short-term physiological effects and long-term clinical benefits of combination therapy in patients with HF across the LVEF spectrum. The corresponding estimates of starting ARNI de novo (upfront initiation) are even greater and shown in Extended Data Figs. 6 and 7.

The expected treatment effects for each therapy and their combinations have been made available in an online calculator (<https://hfmodel.org/>). Example case 1 shows the estimated treatment effects of starting three-drug combination ARNI (de novo) + SGLT2i + sMRA in

Table 1 | Participant characteristics of pivotal randomized trials used to derive the heart failure model

Trial	DAPA-HF	DELIVER	PARADIGM-HF	PARAGON-HF	RALES	TOPCAT-Americas	FINEARTS-HF	CHARM-Alternative	CHARM-Preserved
Comparison	Dapagliflozin versus placebo	Dapagliflozin versus placebo	Sacubitril-valsartan versus enalapril	Sacubitril-valsartan versus valsartan	Spironolactone versus placebo	Spironolactone versus placebo	Finerenone versus placebo	Candesartan versus placebo	Candesartan versus placebo
Population	HFpEF	HFmrEF/HFpEF	HFpEF	HFmrEF/HFpEF	HFpEF	HFmrEF/HFpEF	HFmrEF/HFpEF	HFpEF	HFmrEF/HFpEF
Number of participants	4,744	6,263	8,442	4,822	1,663	1,767	6,001	2,028	3,023
Age (years)	66±11	72±10	64±11	73±8	65±12	72±10	72±10	66±11	67±11
Women	1,109 (23%)	2,747 (46%)	1,832 (22%)	2,479 (52%)	446 (27%)	882 (50%)	2,732 (46%)	646 (32%)	1,212 (40%)
Body mass index (kg m ⁻²)	28±6	30±6	28±6	30±5	NA	34±8	30±6	27±5	29±6
Diabetes	1,983 (42%)	2,806 (45%)	2,907 (35%)	2,062 (43%)	369 (22%)	788 (45%)	2,439 (41%)	545 (27%)	857 (28%)
Atrial fibrillation	1,818 (38%)	3,552 (57%)	3,091 (37%)	2,521 (53%)	NA	743 (42%)	3,273 (55%)	515 (25%)	881 (29%)
Previous hospital admission for heart failure	2,493 (53%)	3,724 (59%)	5274 (63%)	2,306 (48%)	NA	1,040 (59%)	3,619 (60%)	1385 (68%)	2,076 (69%)
SBP (mmHg)	122±16	128±15	121±15	131±15	122±20	128±16	129±15	130±19	136±18
DBP (mmHg)	73±10	74±10	74±10	74±11	75±12	71±11	75±10	77±11	78±11
eGFR (ml min ⁻¹ 1.73 m ⁻²)	66±19	61±19	68±20	63±19	65±23	64±21	62±20	65±24	69±24
Baseline potassium (mmol l ⁻¹)	4.5±0.5	NA	4.5±0.5	4.5±0.5	4.2±0.4	4.2±0.4	4.4±0.5	4.3±0.4	4.4±0.4
LVEF (%)	31±7	54±9	29±6	58±8	25±7	58±8	53±8	30±7	54±9
Baseline medical therapy									
Loop diuretics	4,433 (93%) ^a	4,811 (77%)	6738 (80%)	3,757 (78%)	1,663 (100%)	1,573 (74%) ^a	5,239 (87%)	1,733 (85%) ^a	2,259 (75%)
β-blocker	4,558 (96%)	5,177 (83%)	7,811 (93%)	3,821 (80%)	171 (10%)	1,387 (79%)	5,095 (85%)	1,106 (55%)	1,684 (56%)
ACE inhibitor or ARB	3,952 (83%)	4,543 (73%)	8,379 (100%)	4,139 (86%)	1,589 (96%)	1,395 (79%)	4,758 (79%)	NA	NA
Mineralocorticoid receptor antagonist	3,370 (71%)	2,667 (43%)	4,671 (56%)	1,238 (26%)	NA	NA	NA	483 (24%)	352 (12%)
SGLT2 inhibitor	NA	NA	NA	NA	NA	NA	817 (14%)	NA	NA

NA, not available or applicable. ^aany diuretic use.

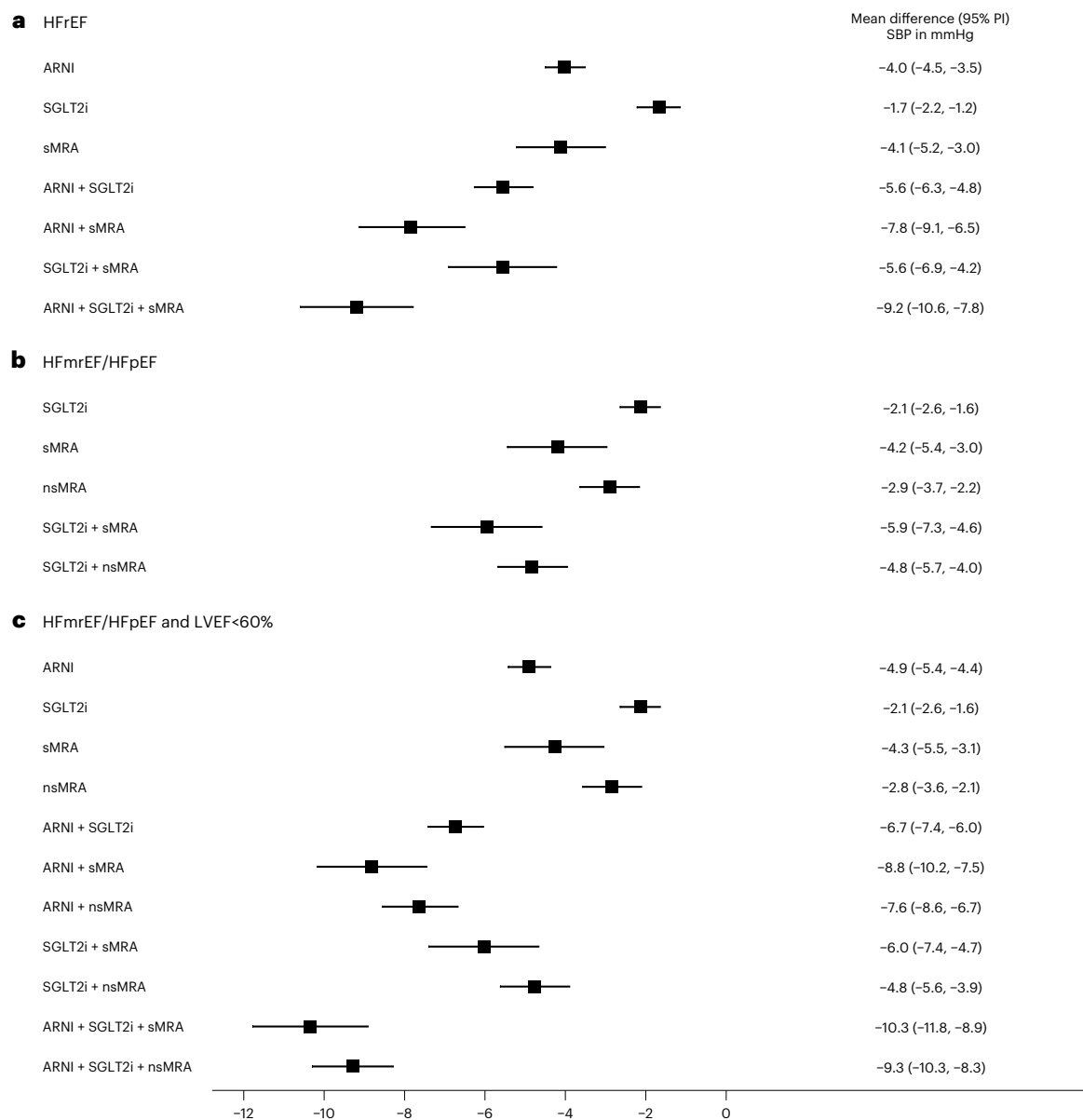


Fig. 1 | Estimated treatment effects of combination comprehensive medical therapy on SBP. a–c, Estimated treatment effects on SBP are shown for patients with HFrEF (**a**), all patients with HFmrEF/HFpEF (**b**) and patients with HFmrEF/HFpEF with LVEF < 60% (**c**). The center points and error bars represent the predicted treatment effects and 95% PIs. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming

additive effects of different drug classes after covariate adjustment. The PIs are calculated with nonparametric bootstrapping using the first-order normal approximation with 300 replicates. **a–c** apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 32,237 participants, with exact sample sizes for each drug class and the full model equations given in the Supplementary Methods.

a 70-year-old man with HFrEF, a prior HF hospitalization, BMI 35 kg m⁻², baseline BP 120/60 mmHg, eGFR 50 ml min⁻¹ m⁻² and baseline serum potassium 4.0 mmol l⁻¹ (Supplementary Fig. 8). The estimated treatment effects on SBP, eGFR and serum potassium would be -12.4 (-14.2, -10.2) mmHg, -10.7 (-12.8, -8.8) and +0.49 (0.41, 0.57), respectively. Meanwhile, the expected relative risk reduction in cardiovascular death and worsening HF events is 73% (65%, 79%). Case 2 shows the estimated treatment effect of adding SGLT2i + nsMRA in a 60-year-old woman with HFpEF, no prior HF hospitalization, BMI 25 kg m⁻², baseline BP 140/80 mmHg, eGFR 40 ml min⁻¹ m⁻² and serum potassium 4.9 mmol l⁻¹. The estimated treatment effect on eGFR and serum potassium is -4.3 (-5.9, -2.8) ml min⁻¹ m⁻² and +0.19 (0.14, 0.23) mmol l⁻¹, respectively, with an estimated relative risk reduction in worsening HF events

of 31% (19%, 41%). Extended Data Table 1 shows a hypothetical framework for using the tool to guide risk-based monitoring for optimization of medical therapies in HF.

Discussion

In this participant-level analysis of 38,753 patients from nine pivotal randomized trials, we quantified individualized treatment effects of comprehensive, disease-modifying HF therapies, alone and in combination, on BP, kidney function and serum potassium. We translated these estimates into an online calculator that provides individualized estimates according to age, sex, body mass index, LVEF, prior HF hospitalization status and baseline BP, kidney function and serum potassium levels (<https://hfmodel.org/>). Across HFrEF and HFmrEF/

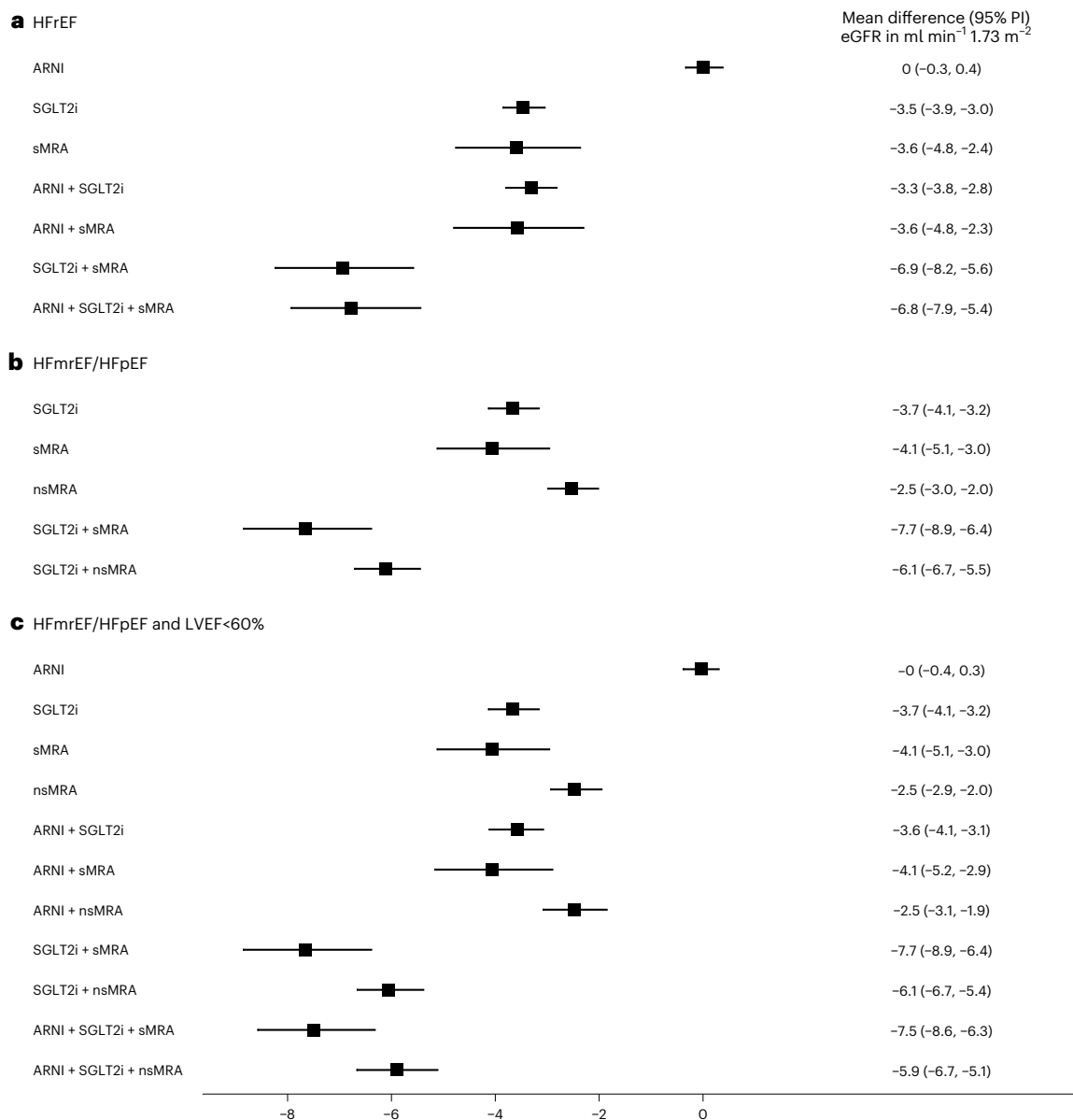


Fig. 2 | Estimated treatment effects of combination comprehensive medical therapy on eGFR. a–c, Estimated treatment effects on eGFR are shown for patients with HFrEF (**a**), all patients with HFmrEF/HFpEF (**b**) and patients with HFmrEF/HFpEF with LVEF < 60% (**c**). The center points and error bars represent the predicted treatment effects and 95% PIs. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming

additive effects of different drug classes after covariate adjustment. The PIs are calculated with nonparametric bootstrapping using the first-order normal approximation with 300 replicates. **a–c** apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 31,525 participants, with exact sample sizes for each drug class and the full model equations given in the Supplementary Methods.

HFpEF populations, comprehensive medical therapy was associated with modest reductions in BP and early declines in eGFR, alongside small to modest increases in serum potassium with some therapies. These modest treatment effects on safety parameters contrast to the large benefits in reductions in cardiovascular death and worsening HF events.

Most patients in contemporary HF practice are treated with limited pharmacotherapies such as an ACEI or ARB and a β -blocker^{2–4}. As such, we focused on early physiological responses with medical therapy optimization, including switching or starting ARNI, MRA (both steroidal and nonsteroidal) and SGLT2i. Although the use of ARNI and steroidal MRA in patients with HFmrEF or HFpEF is variably recommended, we included these therapies to allow clinicians to estimate

their expected physiological effects in patients for whom treatment is being considered. We did not consider other pharmacotherapies such as vericiguat, digoxin, ivabradine, or isosorbide dinitrate/hydralazine given limited support in contemporary guidelines^{8,13}. We provide clinicians with estimated changes in key physiological measures together with estimated risk reductions on worsening HF events with the proposed therapies. This framework may facilitate shared decision-making with patients and caregivers and also guide anticipatory guidance after initiation of therapies. Counseling regarding expected early physiological responses, together with forecasted long-term gains in disease trajectory, may promote long-term persistence and adherence to therapy. Importantly, these estimates of both efficacy and physiological parameters must be interpreted in the context of the trial

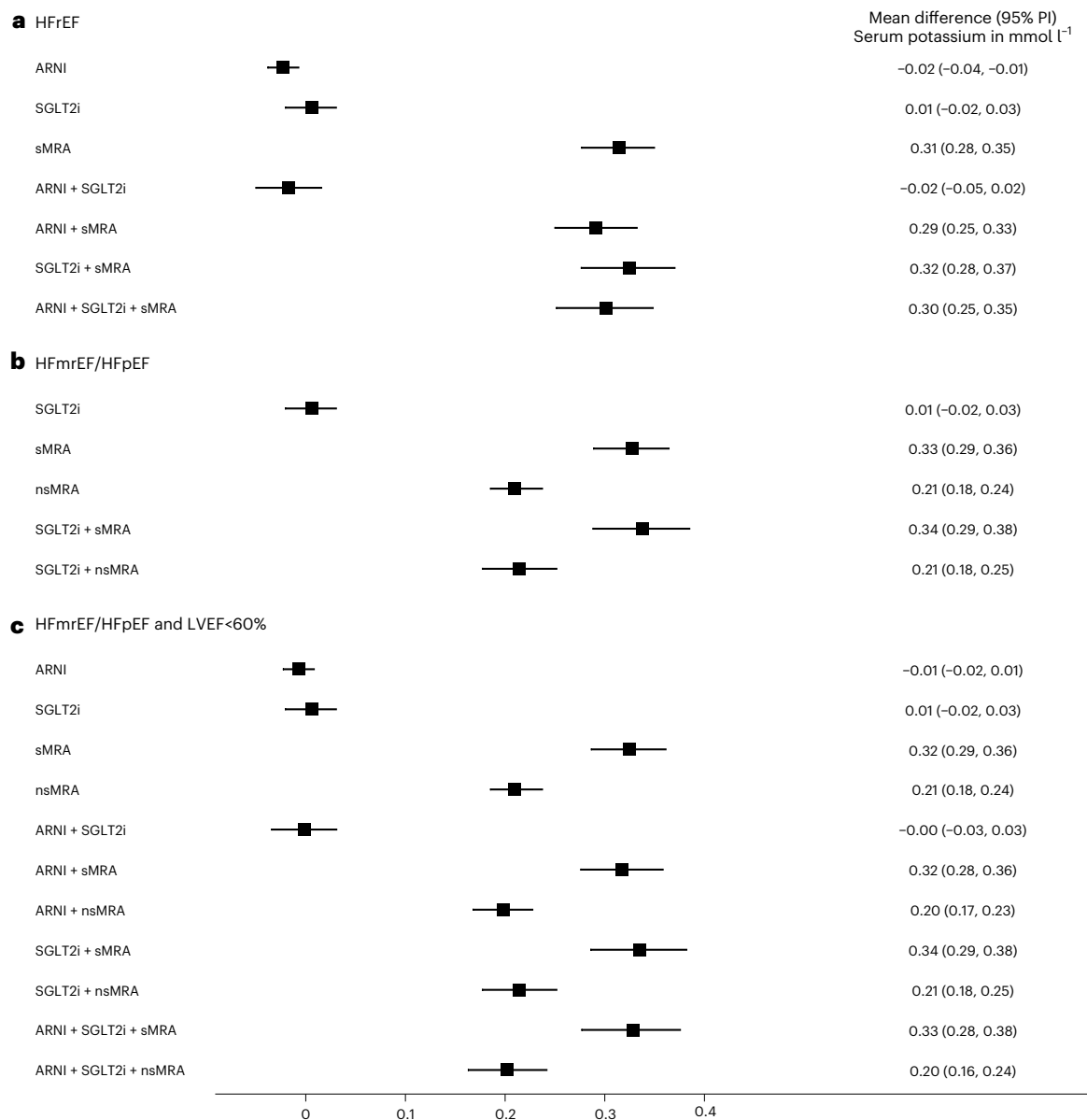


Fig. 3 | Estimated treatment effects of combination comprehensive medical therapy on serum potassium. a–c, Estimated treatment effects on serum potassium are shown for patients with HFrEF (**a**), all patients with HFmrEF/HFpEF (**b**) and patients with HFmrEF/HFpEF with LVEF < 60% (**c**). The center points and error bars represent the predicted treatment effects and 95% PIs. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression

equations, assuming additive effects of different drug classes after covariate adjustment. The PIs are calculated with nonparametric bootstrapping using the first-order normal approximation with 300 replicates. **a–c** apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 25,771 participants, with exact sample sizes for each drug class and the full model equations given in the Supplementary Methods.

population, who met strict eligibility criteria. The observed changes in BP, eGFR and serum potassium were approximately 20% greater than the predicted estimates, which may partly reflect regression to the mean. A prospective study that incorporates these estimates to guide medical therapy implementation is needed.

Fear of adverse events, namely hypotension, kidney dysfunction, and hyperkalemia, is a key driver of therapeutic inertia in HF^{11,12}. The current prescribing paradigm for HF requires titration or optimization of medical therapy as frequently as every 1–2 weeks according to patient-reported symptoms, vital signs, and laboratory findings⁸. However, these physiologic measures have substantial biological variability and measurement error, meaning most of the measured changes from baseline to follow-up are not related to the inciting therapy^{14–17}.

Short-term changes in BP after starting BP-lowering therapies are not reliably associated with true treatment effects on BP¹⁸. Similar patterns have been observed for early declines in kidney function after initiation of inhibitors of renin–angiotensin–aldosterone, where initial eGFR changes are poorly correlated with longer-term renal trajectory or clinical outcomes^{19,20}. The consequence is that early fluctuations in BP, eGFR and serum potassium may be misattributed to drug intolerance, prompting unnecessary dose reduction, discontinuation, or delays in intensification.

Our study provides trial-derived estimates of true short-term treatment effects of therapy on BP, eGFR and serum potassium. Notably, the treatment effects are modest in size, and smaller than the within-person variability for such measures. For example, the treatment effect on

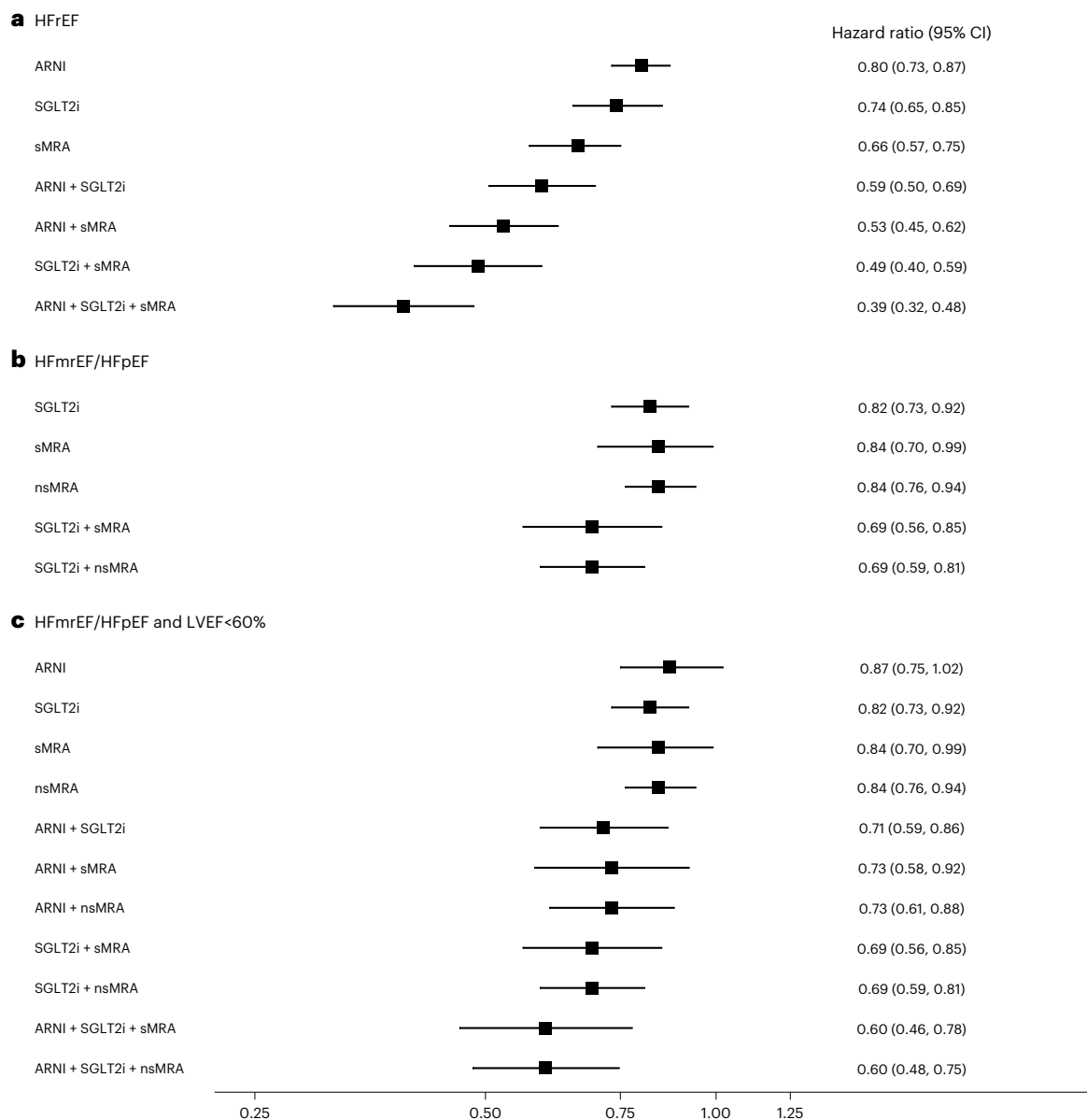


Fig. 4 | Estimated treatment effects of combination therapy in HF on the composite endpoint of cardiovascular death and worsening HF events.

a–c. Estimated treatment effects on the composite endpoint of cardiovascular death and worsening HF events are shown for patients with HFrEF (**a**), all patients with HFmrEF/HFpEF (**b**) and patients with HFmrEF/HFpEF with LVEF < 60% (**c**). The center points and error bars represent the mean hazard ratios and 95% confidence intervals. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. Treatment estimates are derived from 8,442 participants

in PARADIGM-HF, 4,744 participants from DAPA-HF, 1,663 participants in RALES, 4,822 participants in PARAGON-HF; 6,263 participants in DELIVER; 1,767 participants from the Americas cohort of the TOPCAT trial and 6,001 participants from FINEARTS-HF. ARNI, angiotensin receptor blocker – neprilysin inhibitor; CI, confidence interval; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, steroidal mineralocorticoid receptor antagonist; nsMRA, nonsteroidal mineralocorticoid receptor antagonist; SGLT2i, sodium glucose cotransporter 2 inhibitors.

serum potassium for the triple combination of ARNI + SGLT2i + steroidal MRA is +0.30 mmol l⁻¹, which is smaller than the reference change value for serum potassium measurements²¹. Therefore, most of the potassium fluctuations that are observed for an individual are likely related to intrinsic variability and measurement error rather than attributable to treatment. Clinicians should anticipate changes in these safety parameters, and patients in whom the largest changes are predicted are also those receiving the most comprehensive medical therapy. These trial-derived estimates can serve as an anchor against which observed physiological changes can be interpreted in clinical practice. Although heterogeneity in BP response to antihypertensives exist²², we do not currently have the tools to ascertain these relatively

modest differences with adequate precision^{14,15,18,23}. Whether heterogeneity in terms of short-term treatment-related changes in eGFR and serum potassium occurs remains unclear. Adverse effects for HF therapies do exist, including those unrelated to hypotension, kidney dysfunction or hyperkalemia. Indeed, the reported rate of treatment discontinuation due to such adverse effects for ARNI, SGLT2i and MRA is low in the pivotal trials.

Guidelines offer fixed suggested schedules for follow-up clinical and laboratory monitoring, but do not consider an individual patient's risk of developing hypotension or abnormal laboratory values^{8,13}. This calculator offers the potential for tailoring the follow-up protocols to individual risk profiles. However, such a theoretical approach

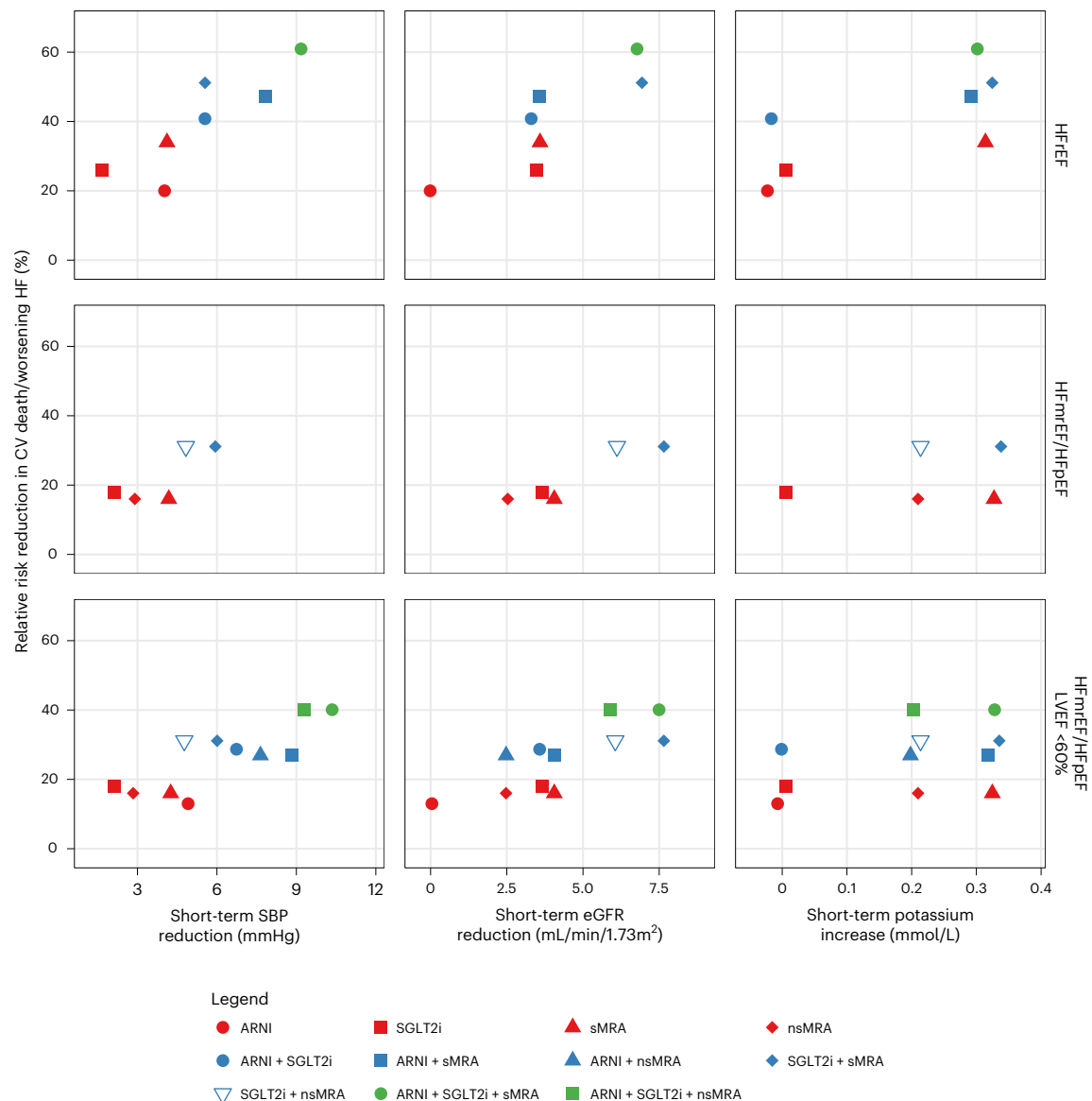


Fig. 5 | Short-term physiological effects and long-term clinical benefit of combination comprehensive medical therapy. This figure compares the risk reductions in cardiovascular death and worsening HF with the short-term changes in SBP, eGFR and potassium using different combinations of medical therapy. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. Each point represents a treatment combination for the specified heart failure population. Center points and horizontal error bars represent the mean treatment effect and 95% PIs for the predicted short-term change in BP, eGFR and serum potassium. Center points and vertical error

bars represent the mean treatment effect and 95% confidence intervals for the relative risk reduction in cardiovascular death or worsening heart failure. Colors denote number of drug classes: red = monotherapy, blue = dual therapy, green = triple therapy. Shapes distinguish individual drug combinations within each therapy group as shown in the legend. Treatment estimates are derived from 8,442 participants in PARADIGM-HF; 4,744 participants from DAPA-HF; 1,663 participants in RALES; 4,822 participants in PARAGON-HF; 6,263 participants in DELIVER; 1,767 participants from the Americas cohort of the TOPCAT trial and 6,001 participants from FINEARTS-HF.

requires prospective testing in randomized implementation trials. Low estimated risk of changes in BP, kidney function and potassium levels may provide reassurance to clinicians and patients and help to overcome clinical inertia. However, patients at high estimated risk of such changes should still be considered for risk-lowering therapies, as these short-term changes should be counter-balanced against long-term potential for clinical benefit in reducing HF events and death. Indeed, even in dedicated combination therapy trials, large early changes in these parameters were reversible (upon drug cessation) and did not usually lead to symptoms, require hospitalization or urgent intervention²⁴.

These estimates were derived from trial doses of each therapy, and we do not provide dose-specific estimates. Nevertheless, the guidelines recommend uptitration of HF therapies to doses tested

in trials⁸. Clinicians should also consider important inclusion and exclusion criteria of the trials when using these therapies, and these estimates may not be generalizable to broader risk populations. For example, patients with a serum potassium >5 mmol l⁻¹ or severe kidney impairment were generally excluded from trials of MRA and ARNI. The ARNI trials also included a run-in period, which would have excluded patients who were unable to tolerate the treatment. Although we adjusted for differences in trial design in the regression models, ascertainment of measures varied between trials, including different timepoints for outcome assessment and different equations to calculate eGFR. Our analysis focused on short-term (2–12 weeks) changes in BP, eGFR and serum potassium, corresponding to the period in which the physiological effects of each therapy are expected to be most apparent. The impact of these differences in timepoint on the

estimated treatment effects is likely small. The assumption that the treatment effects of each therapy are additive is supported by several lines of evidence—first, each of the evaluated therapies has a distinct mechanism of action with no known pharmacological interactions. Second, we found the treatment effects of each drug class were consistent across different background therapies. Third, we performed validation analyses on external combination trials specifically testing combination therapies. Fourth, these combination trials have also separately concluded that the effects of combined SGLT2i and MRAs on BP, kidney function, and serum potassium in patients with HFpEF and chronic kidney disease are additive^{24,25}. The observed changes in BP, eGFR and serum potassium were about 20% greater than the calculated estimates, which may reflect some degree of regression to the mean, particularly for SBP^{26,27}. In clinical practice, the true treatment response in an individual patient is likely to lie between the observed change after treatment initiation and the expected average treatment effect estimated from randomized trials. Our regression models for ARB were developed using the CHARM program, where serum potassium/eGFR levels were only available in a subgroup of the participants, which may have underpowered our ability to identify effect modifiers. The results from TOPCAT may need to be interpreted with caution because participants from Russia and Georgia were excluded, as previously documented^{28,29}.

Among patients with HF, we provide individualized estimates of the treatment effect from combination therapy on short-term changes in BP, kidney function and serum potassium. The expected treatment effects on these safety parameters are modest and contrast to the large benefits in reductions in cardiovascular death and worsening HF events. These estimates are available in an online calculator which may facilitate treatment decisions around the use of combination therapy in HF.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-026-04623-z>.

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Methods

Study design

This was an individual participant-level analysis of pivotal trials that supported the regulatory evaluation of comprehensive disease-modifying pharmacological therapies approved by the US Food and Drug Administration for the management of HF. For each comprehensive disease-modifying drug class, we included a pivotal trial in each HF population (HFrEF and HFmrEF/HFpEF). The study was registered on PROSPERO CRD420261299819. We estimated the treatment effects of ARNI, SGLT2i, sMRA and nsMRA on the short-term (2–12 weeks) changes in BP, eGFR and serum potassium from baseline to the first follow-up visit after dose titration (Extended Data Table 2). The included trials investigated the efficacy and safety of SGLT2i, ARNI and sMRA in patients with HFrEF and SGLT2i, ARNI, sMRA or nsMRA in patients with HFmrEF/HFpEF. Finally, the treatment effects of ARB were also estimated to calculate the treatment effects of ARNI versus a putative placebo. All the included trials were approved by ethics committees or institutional review boards at all participating sites and all participants provided written informed consent. Ethics oversight was obtained for each individual trial and has been published separately.

Dapagliflozin and Prevention of Adverse-outcomes in Heart Failure (DAPA-HF)

From 2017 to 2018, the DAPA-HF trial randomly assigned 4,744 patients aged 18 years or older who had symptomatic HF with LVEF $\leq 40\%$, and elevated natriuretic peptides to dapagliflozin 10 mg once daily or matching placebo³⁰. Patients with symptomatic hypotension or SBP < 95 mmHg on two consecutive measurements and eGFR < 30 ml min⁻¹ m⁻² were excluded. Background therapy included β -blockers, ACEIs/ARBs/ARNIs and MRAs. We assessed the change in BP, eGFR and serum potassium from baseline to 2 weeks. Median follow-up was 18 months.

DELIVER

From 2018 to 2021, the DELIVER trial randomly assigned 6,263 adults ≥ 40 years with symptomatic HF and an LVEF $> 40\%$ to dapagliflozin 10 mg once daily or matching placebo³¹. All participants were required to have evidence of structural heart disease and elevated natriuretic peptide levels. Patients with symptomatic hypotension or SBP < 95 mmHg on two consecutive measurements and eGFR < 25 ml min⁻¹ m⁻² were excluded. At the time, there were no standard disease-modifying therapies available. We assessed the change in BP and eGFR from baseline to 4 weeks; post-baseline potassium levels were not collected in DELIVER because DAPA-HF and the totality of evidence from prior trials had demonstrated SGLT2i had little impact on serum potassium levels. Median follow-up was 2.3 years.

Prospective Comparison of ARNI with ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure Trial (PARADIGM-HF)

From 2009 to 2012, the PARADIGM-HF randomized 8,442 patients aged 18 years or older who had symptomatic HF and LVEF $\leq 40\%$ (changed to $\leq 35\%$ by protocol amendment), and elevated levels of natriuretic peptides to enalapril 10 mg twice daily or sacubitril/valsartan 97/103 mg twice daily³². Participants with eGFR < 30 ml min⁻¹ m⁻², symptomatic hypotension or SBP < 95 mmHg and serum potassium > 5.4 mmol l⁻¹ at randomization were excluded. Background therapy included β -blockers, ACEIs/ARBs and MRAs. All participants had tolerated target doses of both study drugs during single-blind run-in periods before randomization. We assessed the change in BP, eGFR and serum potassium from baseline to 2 weeks. Median follow-up was 27 months.

Prospective Comparison of ARNI with ARB Global Outcomes in HF with Preserved Ejection Fraction (PARAGON-HF)

From 2014 to 2016, the PARAGON-HF trial randomly assigned 4,822 adults ≥ 18 years with symptomatic HF, LVEF $\geq 45\%$, structural heart

disease, and elevated natriuretic peptides to sacubitril-valsartan 97/103 mg twice daily versus valsartan 160 mg twice daily³³. Participants with SBP < 110 or symptomatic hypotension, eGFR < 25 ml min⁻¹ m⁻² and serum potassium > 5.4 mmol l⁻¹ at randomization were excluded. At the time, there were no standard disease-modifying therapies available. All participants had tolerated half-target doses of both study drugs before randomization. We assessed the change in BP, eGFR and serum potassium from baseline to 4 weeks. Median follow-up was 2.9 years.

Randomized Aldactone Evaluation Study (RALES)

From 1994 to 1999, the RALES randomized 822 patients with severe HF with LVEF $\leq 35\%$, New York Heart Association class III–IV to spironolactone 25 mg or matching placebo³⁴. Participants with serum creatinine > 2.5 mg dl⁻¹ (225 μ mol l⁻¹) and serum potassium > 5.0 mmol l⁻¹ were excluded. Background therapy included ACEIs/ARBs and low rates of β -blockers. The dose of spironolactone was increased to 50 mg at 8 weeks if the patient showed signs or symptoms of progression of HF. We assessed the change in BP, eGFR and potassium from baseline to 12 weeks. Mean follow-up was 2 years.

TOPCAT Americas

From 2006 to 2012, 3,445 participants aged ≥ 50 years with HF and LVEF $\geq 45\%$ with a prior history of hospitalization for HF in the prior 12 months or elevated natriuretic peptide levels were randomized to spironolactone 15 mg or matching placebo³⁵. Participants also needed to have controlled BP defined as systolic BP (SBP) ≤ 140 mmHg or ≤ 160 mmHg on three or more antihypertensives, and a serum potassium level < 5.0 mmol l⁻¹. Spironolactone dose was increased to 30 mg at 4 weeks. We assessed the change in BP, eGFR and serum potassium from baseline to 8 weeks. For this study, we restricted our analyses to the subgroup of participants who were enrolled in the Americas due to concerns that participants enrolled in Russia and Georgia did not have symptomatic HF and may not have been taking study medications^{28,29}. Mean follow-up for participants randomized in the Americas was 2.9 years.

FINEARTS-HF

From 2020 to 2023, the FINEARTS-HF trial randomly assigned 6,001 adults ≥ 40 years with symptomatic HF and an LVEF $\geq 40\%$ to finerenone or matching placebo³⁶. All participants were required to have evidence of structural heart disease and elevated natriuretic peptide levels. Participants with symptomatic hypotension with SBP < 90 mmHg, serum potassium levels > 5.0 mmol l⁻¹ or eGFR < 25 ml min⁻¹ 1.73 m⁻² were excluded. There were low rates of SGLT2i use at the time of the trial. The starting dose of finerenone was 10 mg for those with baseline eGFR ≤ 60 ml min⁻¹ m⁻² and 20 mg for baseline eGFR > 60 ml min⁻¹ m⁻². At 1 month, the dose of finerenone should be increased to 20 mg (for those patients with baseline eGFR ≤ 60 ml min⁻¹ m⁻²) or 40 mg (for those patients with baseline eGFR > 60 ml min⁻¹ m⁻²) if K < 5.0 and eGFR decrease was less than 30% at 1 month. We assessed the change in BP, eGFR and potassium from baseline to 4 weeks, as dose changes that occurred after 4 weeks were dependent on the changes in eGFR. Median follow-up was 2.7 years.

CHARM-Alternative

From 1999 to 2001, the Candesartan in Heart Failure—Assessment of Reduction in Mortality and Morbidity (CHARM)-Alternative trial randomized 2,028 patients with symptomatic HF, LVEF $\leq 40\%$ who were intolerant to ACEIs to candesartan or matching placebo to a target dose of 32 mg by 6 weeks^{37,38}. We assessed the change in BP, GFR and potassium from baseline to 6 weeks. Median follow-up was 2.8 years.

CHARM-Preserved

From 1999 to 2000, the CHARM-Preserved trial randomly assigned 3,023 patients with symptomatic HF, LVEF $\geq 40\%$ and history of a cardiac hospitalization to candesartan or matching placebo to a target dose of

32 mg by 6 weeks³⁸. We assessed the change in BP, GFR and potassium from baseline to 6 weeks. Median follow-up was 3.1 years.

Clinical outcomes

Key measures evaluated were changes in SBP, DBP, eGFR and serum potassium levels. Additionally, we assessed the composite endpoint of cardiovascular death and first worsening HF event to estimate the expected treatment benefits from combination therapy.

Statistical analyses

We pooled participant-level data by randomized drug class under the rationale that such trials estimate a common class-level treatment effect on the safety parameters. We constructed separate linear regression equations for each therapy and clinical outcome (SBP, DBP, eGFR and serum potassium). We considered the following prespecified list of covariates for each regression model: trial ID, baseline LVEF, baseline BMI, baseline SBP or DBP, baseline eGFR, baseline serum potassium levels, age, sex, prior HF hospitalization, baseline use of β -blocker, baseline use of ACEI or ARB, baseline use of ARNI, baseline use of MRA or baseline use of SGLT2i where appropriate. We used a complete case analysis rather than multiple imputation because the number of within-trial missing variables was small and there was no baseline imbalance in characteristics within the trials (Supplementary Table 3). In the CHARM program, laboratory measures were collected in a prespecified subset of patients. For continuous variables that were potential effect modifiers, we tested the relationship between the continuous variables and the treatment effect using restricted cubic splines, choosing the number of knots with the lowest Akaike information criterion. For continuous effect modifiers where the restricted cubic spline indicated a nonlinear treatment-by-covariate relationship, we re-expressed the functional form as a continuous piecewise linear spline, placing knots at the data-identified inflection point rounded to the nearest clinically interpretable value (5 mmHg for BP; 0.5 mmol l⁻¹ for serum potassium and eGFR thresholds corresponding to established chronic kidney disease staging boundaries; Supplementary Fig. 9). We also tested the model fit between the linear piecewise and restricted cubic spline specifications in terms of Akaike information criterion and Bayesian information criterion (Supplementary Table 4). With the exception of trial ID, we only included effect modifiers in the final model for variables that were significant in their covariate-by-treatment interaction term, with a two-tailed $P < 0.05$ as the threshold for significance. Trial ID was included as a mandatory categorical covariate in the final regression model to account for differences in trial protocol and timing of follow-up measures.

To calculate combination treatment effects, we used a previously established method that assumes the treatment effects of each therapy are additive after accounting for the treatment effect of the concurrent therapies on any effect modifiers³⁹. For example, if baseline SBP was an effect modifier for drug A, we adjusted for a lower baseline SBP when calculating the treatment effect of drug B, taking an average of the different permutations of treatment sequence (drug A followed by drug B, and drug B followed by drug A). The full regression equations for calculating the treatment effect for any possible combination can be found in the Supplement. For the main analysis, we estimated the treatment effects of switching from ACEI/ARB to ARNI because PARADIGM-HF and PARAGON-HF were active-controlled trials comparing ARNI versus ACEI/ARB. For all other trials, the comparator group was placebo. As a secondary analysis, we also estimated the treatment effects of starting ARNI 'de novo', by comparing its treatment effects to a putative placebo. Using established methods, we estimated the treatment effect of ARNI versus placebo by adding the treatment effects of ARNI versus ACEI/ARB with the treatment effects of ARB versus placebo from a pooled analysis of the CHARM-Alternative and CHARM-Preserved trials³⁹. We acknowledge upfront that the ARB versus placebo estimates used to anchor these de novo ARNI calculations are derived from the CHARM trials conducted during an

era with markedly lower background medical therapy use, and that extrapolating them to a contemporary treatment context assumes the relative treatment effect of ARB versus placebo is preserved across these differing backgrounds.

We validated the model using four trials that tested combination therapies of the treatments of interest, INITIATE, SOGALDI-PEF, CONFIDENCE and ROTATE-3 (Extended Data Table 3)^{24,25,40,41}. INITIATE and SOGALDI-PEF enrolled populations of patients with heart failure, while CONFIDENCE and ROTATE-3 enrolled patients with chronic kidney disease. We contacted authors and obtained individual participant-level data for the two HF trials^{25,41}. For CONFIDENCE and ROTATE-3 trials, we used trial-level averages as inputs. Using our derived regression equations, we estimated the changes in BP, eGFR and serum potassium for each randomized treatment group. We compared these estimated treatment effects to the observed changes in each parameter at the first follow-up timepoint after treatment was established. For crossover trials (SOGALDI-PEF and ROTATE-3), each treatment period was considered a separate treatment group. For INITIATE, participants randomized to sequential treatment with ARNI and SGLT2i were assessed at both 4 weeks (first visit after monotherapy was established) and at 24 weeks (first visit after combination therapy was established). The consistency between observed and expected efficacy of combination therapies was tested using the root mean square error, mean absolute error and slope of the line of best fit.

We estimated the mean and 95% PIs for one, two-drug and three-drug combinations of comprehensive medical therapy for the following three separate populations: (1) HFref, where indicated therapies include ARNI, SGLT2i and sMRA; (2) HFmrEF/HFpEF, where indicated therapies include SGLT2i and sMRA or nsMRA and (3) HFmrEF/HFpEF with LVEF below 60%, where indicated therapies include ARNI, SGLT2i and sMRA or nsMRA. We used the pooled average covariate profile of each population as inputs in the prediction model. The PIs were calculated with nonparametric bootstrapping using the first-order normal approximation with 300 replicates. For the online calculator, we used 100 replicates for a faster runtime. A full list of all equations used in the calculation of combination treatment effects can be found in the Supplementary Methods.

For reference, to inform overall clinician risk-benefit calculus, we also estimated the treatment effects on the standardized composite endpoint of cardiovascular death and first worsening HF event to assess for benefits of combination therapy. We used individual treatment effects observed in each component trial and calculated the treatment effects of combination therapy using established methods of indirect comparisons, which are commonly applied in putative placebo assessments^{42–44}. We calculated the treatment effect of ARB using estimates from CHARM-Reduced and CHARM-Preserved for HFref and HFmrEF/HFpEF, respectively. All analyses were performed with Stata version 18.1 and R version 4.1.2 (refs. 45,46).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All trial funders are committed to sharing access to patient-level data and supporting clinical documents from eligible studies. Trial data availability is subject to the separate criteria and processes of AstraZeneca ([https://astrazenecagrouptrials.pharmacm.com/ST/](https://astrazenecagrouptrials.pharmacm.com/ST/Submission/) Submission/ Disclosure), Bayer (<https://vivli.org/ourmember/bayer/>), Novartis (<https://www.novartis.com/sites/novartis.com/files/clinical-trial-data-transparency.pdf>) and Pfizer (<https://www.pfizer.com/science/clinical-trials/data-and-results>).

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Author contributions

N.W. and M.V. conceived and designed the study, acquired and interpreted the data, and drafted the manuscript. N.W. and B.L.C. contributed to the study design and analyzed and interpreted the data. M.A.P., J.P.F., F.Z., B.P., M.P., K.F.D., J.J.V.M. and S.D.S. acquired and interpreted the data. B.L.C. managed the data. N.W., B.L.C., M.V. and J.P.F. interpreted the data. All authors critically reviewed the manuscript.

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Competing interests

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Extended Data Table 1 | Hypothetical framework for risk-based monitoring for optimization of medical therapies in heart failure

Risk category	Estimated post-initiation SBP (mmHg)	Estimated post-initiation eGFR (mL/min/1.73m ²)	Estimated post-initiation serum potassium (mmol/L)	Theoretical approach to risk-based initiation and monitoring (requires prospective validation)
Low risk	≥100 mmHg	≥30 mL/min/m ²	<5 mmol/L	• Simultaneous or rapid-sequence initiation of combination therapy. • No routine interim BP or laboratory testing before 4 weeks provided patient is clinically stable and asymptomatic
Moderate risk	90-99 mmHg	20-30 mL/min/m ²	5-5.4 mmol/L	• Simultaneous or rapid-sequence initiation of combination therapy. • Repeat BP and laboratory testing at ~2 weeks or earlier if symptoms develop or clinical status changes
High risk	<90 mmHg	<20 mL/min/m ²	≥5.5 mmol/L	• Consider staggered initiation and uptitration of combination therapy. Close monitoring with repeat BP and laboratory testing within 1-2 weeks (earlier if symptoms develop or clinical status changes)

BP, blood pressure; eGFR, estimated glomerular filtration rate.

Extended Data Table 2 | Timepoints of assessment of blood pressure, GFR and potassium and definitions of worsening heart failure event used for each trial

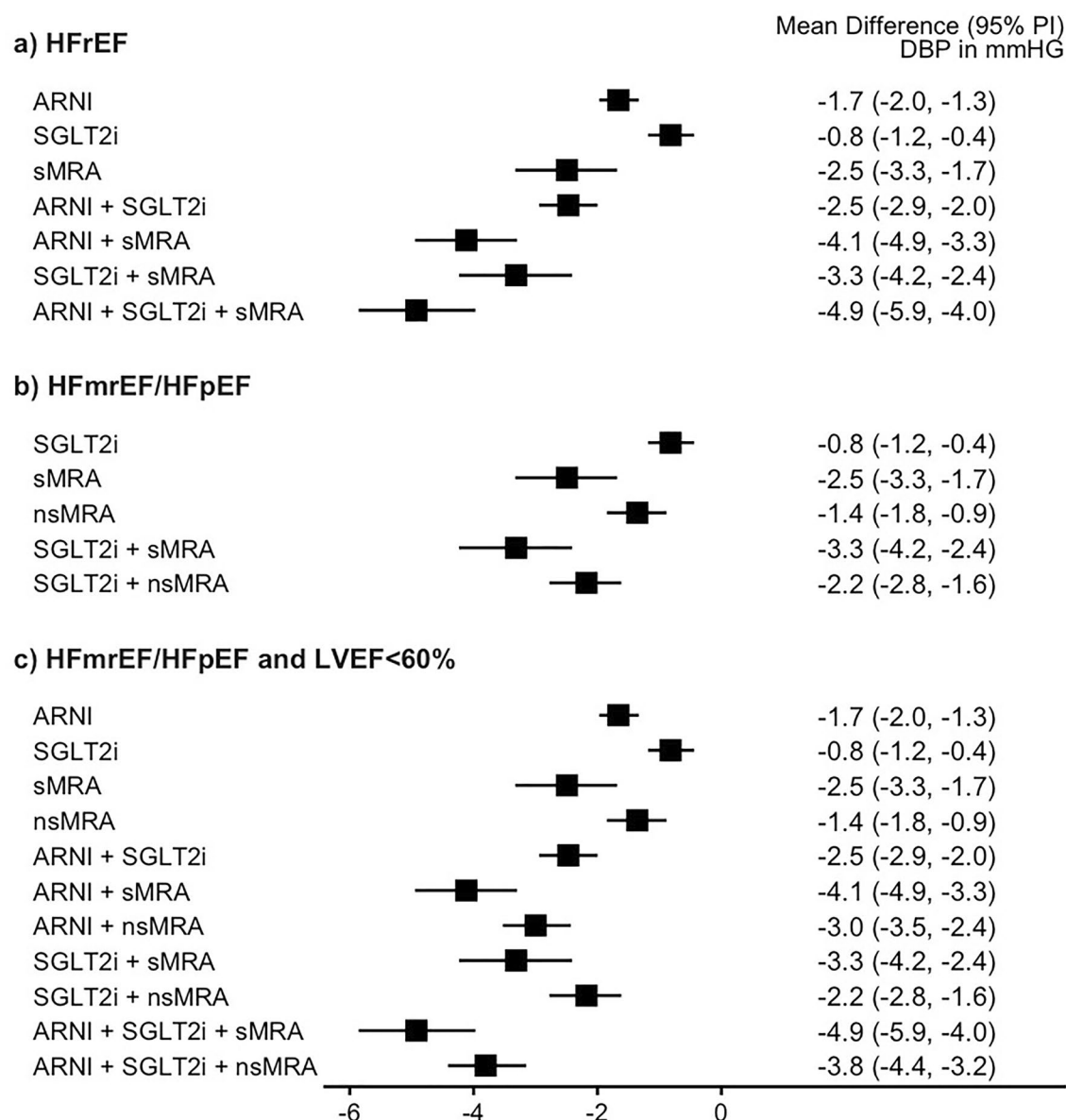
Trial	Active treatment and target dose at timepoint assessment	Timepoint	Physiologic Endpoints assessed	Definition of worsening HF
DAPA-HF	Dapagliflozin 10mg once daily	2 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization or an urgent visit resulting in intravenous therapy for HF
DELIVER	Dapagliflozin 10mg once daily	4 weeks	Systolic BP, Diastolic BP, eGFR	HF hospitalization or an urgent visit resulting in intravenous therapy for HF
PARADIGM-HF	Sacubitril 97mg and valsartan 103mg twice daily	2 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization
PARAGON-HF	Sacubitril 97mg and valsartan 103mg twice daily	4 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization
RALES	Spironolactone 50 mg once daily	12 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization
TOPCAT-Americas	Spironolactone 30 mg once daily	8 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization
FINEARTS-HF	Finerenone 10 mg (GFR ≤ 60 ml/min/m ²) or 20 mg (GFR >60 ml/min/m ²)	4 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization or an urgent visit resulting in intravenous therapy for HF
CHARM-Alternative	Candesartan 32 mg	6 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization
CHARM-Preserved	Candesartan 32 mg	6 weeks	Systolic BP, diastolic BP, eGFR, potassium	HF hospitalization

BP, blood pressure; eGFR, estimated glomerular filtration rate; HF, heart failure.

Extended Data Table 3 | Characteristics of trials included in the validation cohort

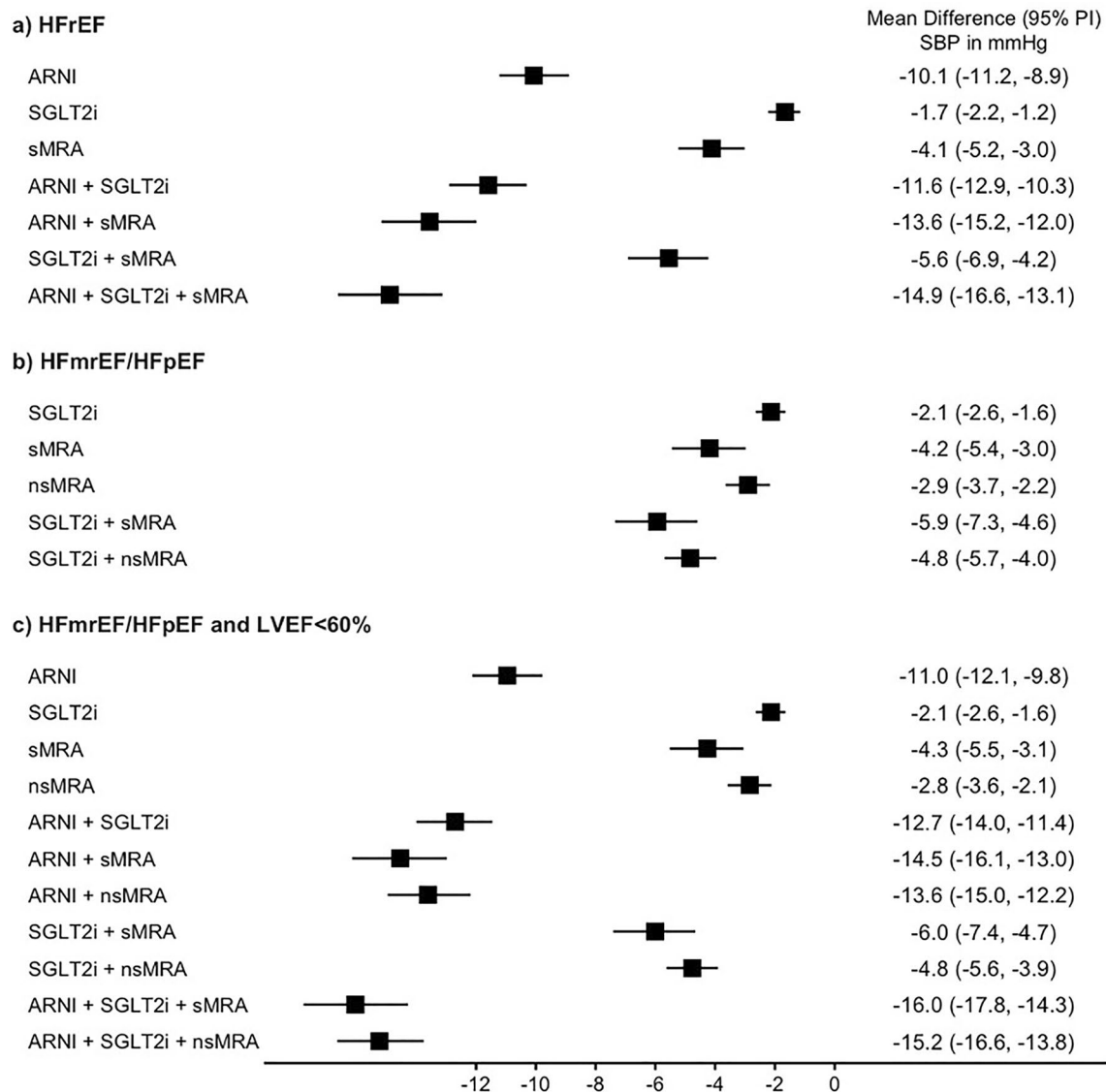
Trial name	INITIATE	SOGALDI-PEF	CONFIDENCE	ROTATE-3
Individual patient data	Yes	Yes	No	No
Number of participants	62	108	800	46
Population	HFrEF	HFmrEF/HFpEF	Chronic kidney disease with type 2 diabetes	Chronic kidney disease
Study design	Open-label RCT	Open-label, cross-over RCT	Double-blind RCT	Open-label, cross-over RCT
Randomized treatment groups	1) ARNI; 2) SGLT2i; 3) ARNI + SGLT2i	1) SGLT2i; 2) MRA; 3) SGLT2i + MRA	1) SGLT2i; 2) nsMRA; 3) nsMRA + SGLT2i	1) SGLT2i, 2) MRA, 3) SGLT2i + MRA
Outcomes assessed	Systolic BP, Diastolic BP, eGFR, serum potassium	Systolic BP, Diastolic BP, eGFR, serum potassium	Systolic BP, eGFR, serum potassium	Systolic BP, eGFR, serum potassium
Follow-up timepoint for outcome assessment	4 and 12 weeks	12 weeks	2 weeks	4 weeks

ARNI, angiotensin receptor blocker – neprilysin inhibitor; BP, blood pressure; eGFR, estimated glomerular filtration rate; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; SGLT2i, sodium glucose cotransporter 2 inhibitors.



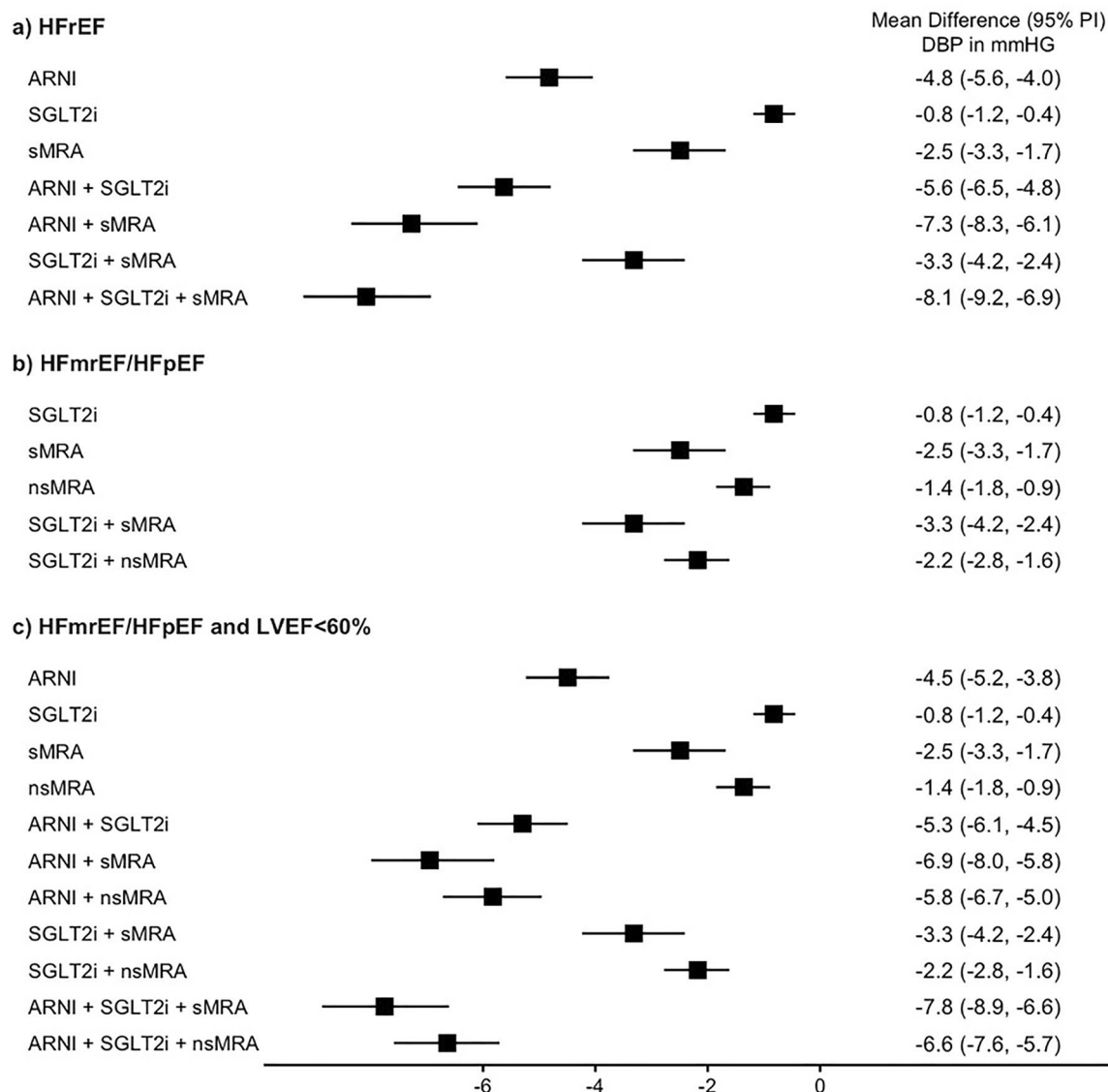
Extended Data Fig. 1 | Estimated treatment effects of combination comprehensive medical therapy on diastolic blood pressure. (a–c) Estimated treatment effects on diastolic blood pressure are shown for patients with HFrEF (a); all patients with HFmrEF/HFpEF (b) and patients with HFmrEF/HFpEF with LVEF < 60% (c). The center points and error bars represent the predicted treatment effects and 95% prediction intervals. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming additive effects of different drug classes after covariate adjustment. The prediction intervals are calculated with non-parametric bootstrapping using the

first-order normal approximation with 300 replicates. a–c apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 32,347 participants, with exact sample sizes for each drug class and the full model equations given in the Supplement. ARNI, angiotensin receptor blocker – neprilysin inhibitor; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, steroidal mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; PI, prediction interval; SBP, systolic blood pressure; SGLT2i, sodium glucose cotransporter 2 inhibitors.



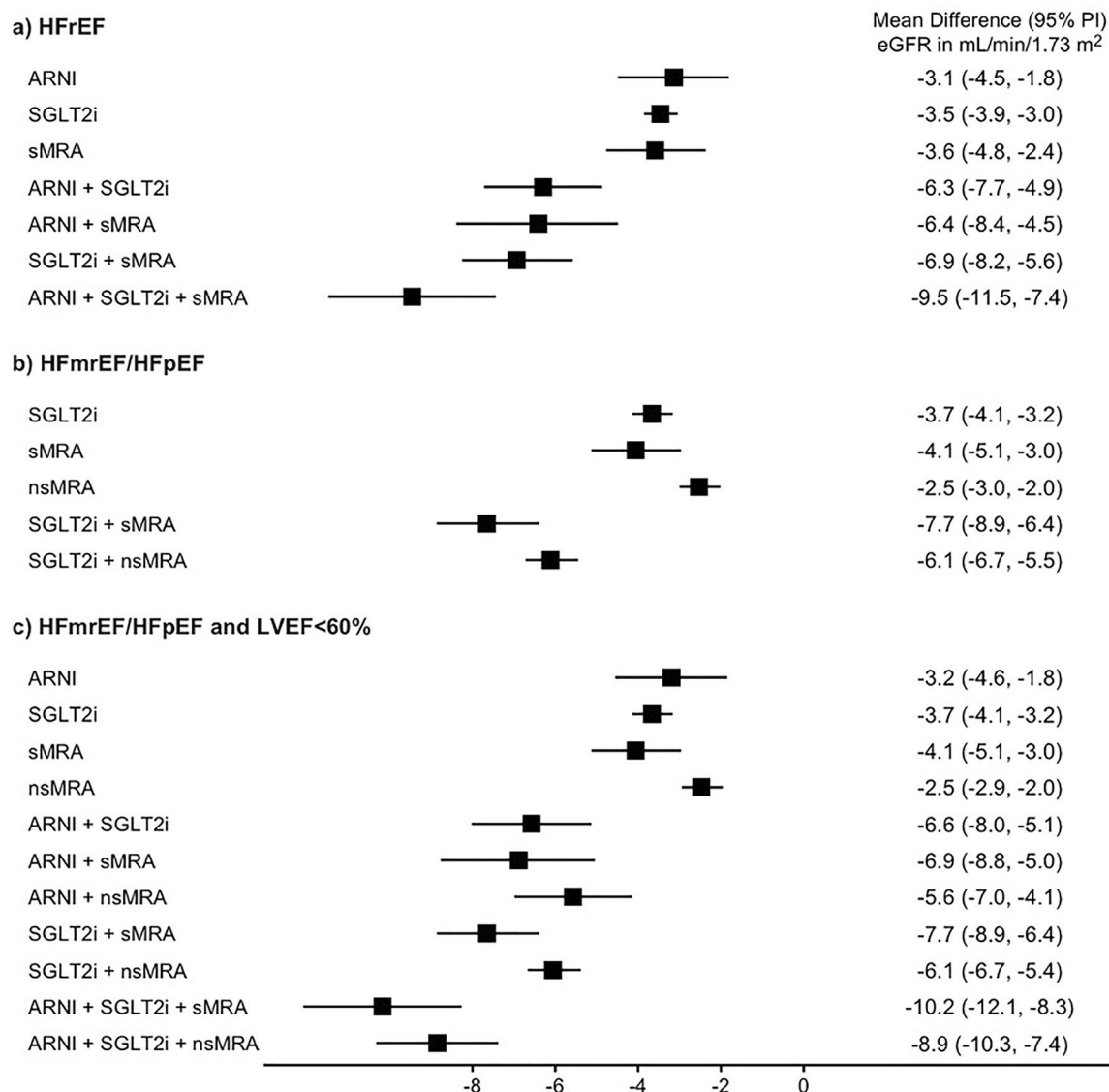
Extended Data Fig. 2 | Estimated treatment effects of combination comprehensive medical therapy with ARNI commenced de novo on systolic blood pressure. (a-c) Estimated treatment effects on systolic blood pressure are shown for patients with HFrEF (a); all patients with HFmrEF/HFpEF (b) and patients with HFmrEF/HFpEF with LVEF < 60% (c). The center points and error bars represent the predicted treatment effects and 95% prediction intervals. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming additive effects of different drug classes after covariate adjustment. The prediction intervals are calculated with non-parametric

bootstrapping using the first-order normal approximation with 300 replicates. **a-c** apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 37,104 participants, with exact sample sizes for each drug class and the full model equations given in the Supplement. ARNI, angiotensin receptor blocker – neprilysin inhibitor; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, steroidal mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; PI, prediction interval; SBP, systolic blood pressure; SGLT2i, sodium glucose cotransporter 2 inhibitors.



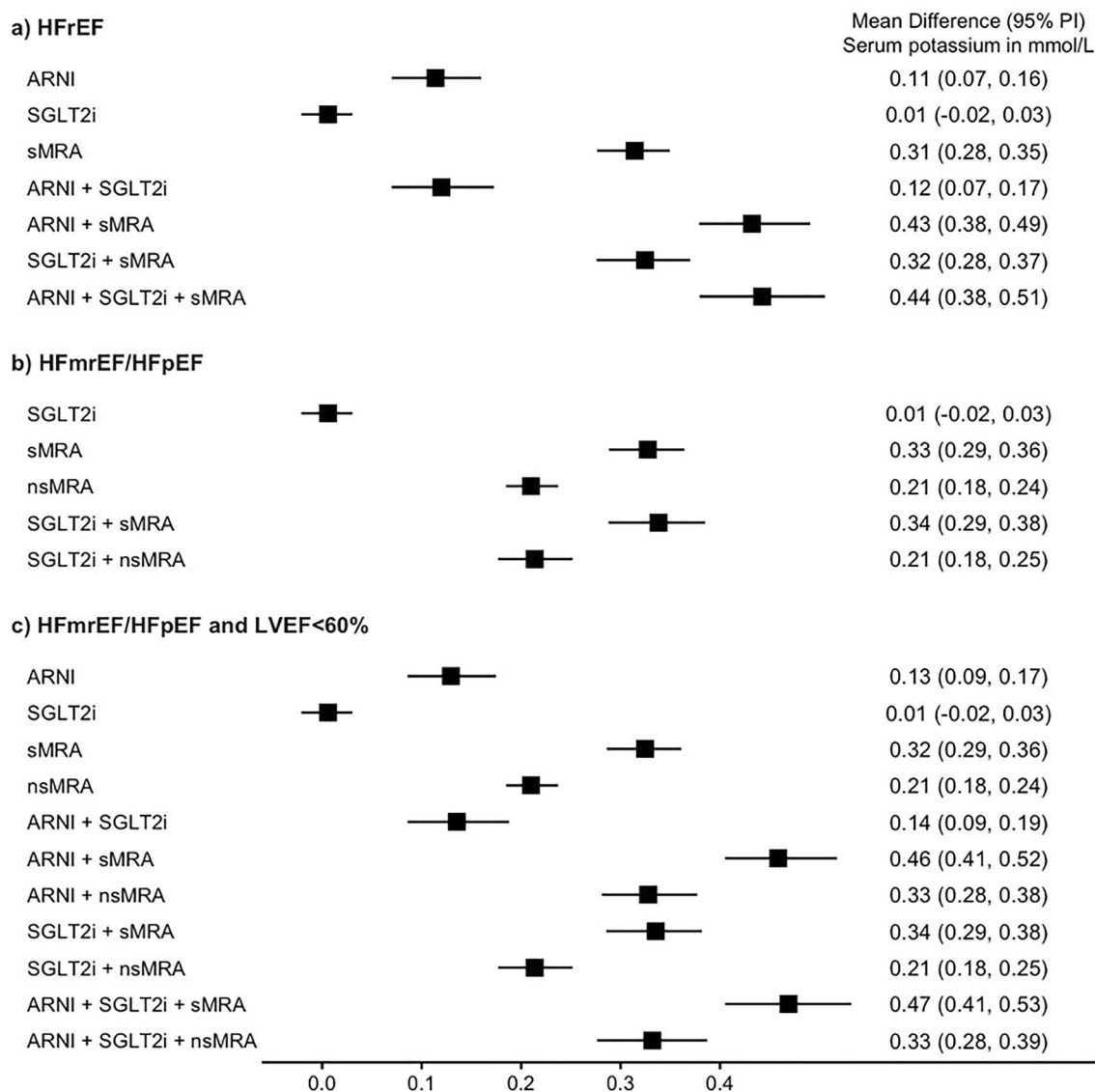
Extended Data Fig. 3 | Estimated treatment effects of combination comprehensive medical therapy with ARNI commenced de novo on diastolic blood pressure. (a-c) Estimated treatment effects on diastolic blood pressure are shown for patients with HFrEF (a); all patients with HFmrEF/HFpEF (b) and patients with HFmrEF/HFpEF with LVEF < 60% (c). The center points and error bars represent the predicted treatment effects and 95% prediction intervals. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming additive effects of different drug classes after covariate adjustment. The prediction intervals are calculated with non-parametric

bootstrapping using the first-order normal approximation with 300 replicates. a-c apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 37,213 participants, with exact sample sizes for each drug class and the full model equations given in the Supplement. ARNI, angiotensin receptor blocker – neprilysin inhibitor; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, steroidal mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; PI, prediction interval; SBP, systolic blood pressure; SGLT2i, sodium glucose cotransporter 2 inhibitors.



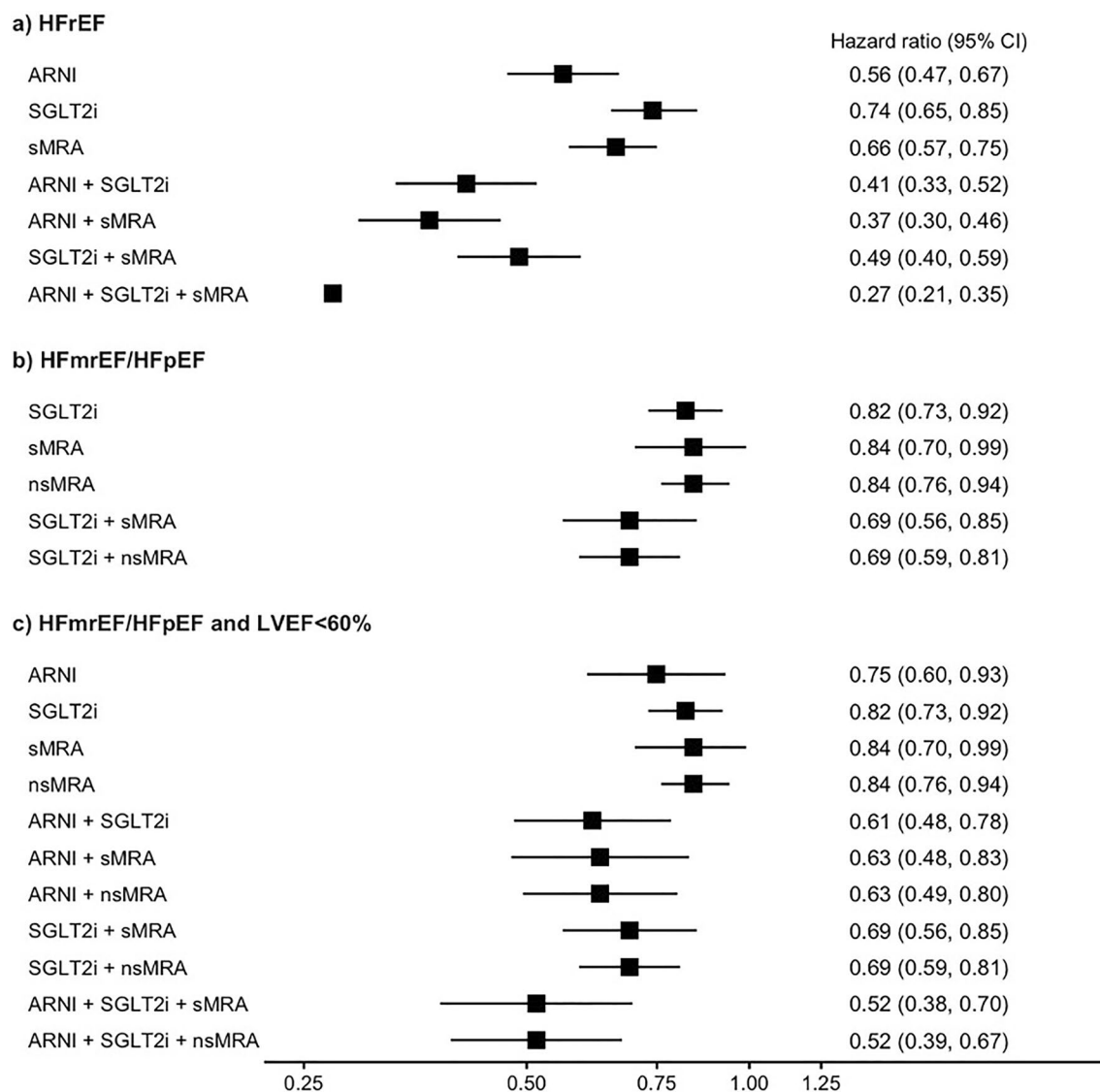
Extended Data Fig. 4 | Estimated treatment effects of combination comprehensive medical therapy with ARNI commenced de novo on eGFR. (a–c) Estimated treatment effects on eGFR are shown for patients with HFrEF (a); all patients with HFmrEF/HFpEF (b) and patients with HFmrEF/HFpEF with LVEF < 60% (c). The center points and error bars represent the predicted treatment effects and 95% prediction intervals. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming additive effects of different drug classes after covariate adjustment. The prediction intervals are calculated with non-parametric bootstrapping using the

first-order normal approximation with 300 replicates. a–c apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 33,092 participants, with exact sample sizes for each drug class and the full model equations given in the Supplement. ARNI, angiotensin receptor blocker – neprilysin inhibitor; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, steroidal mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; PI, prediction interval; SBP, systolic blood pressure; SGLT2i, sodium glucose cotransporter 2 inhibitors.



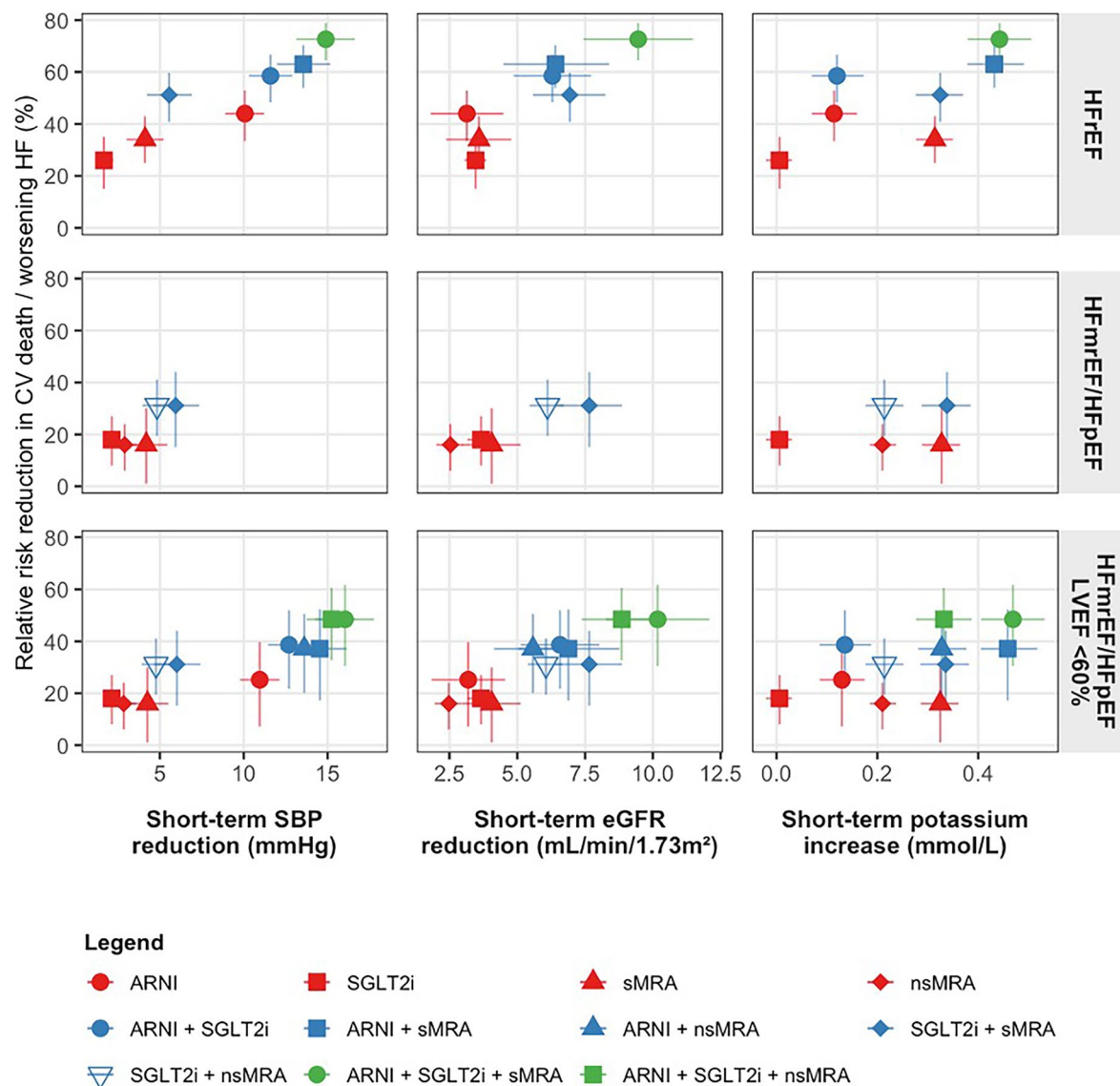
Extended Data Fig. 5 | Estimated treatment effects of combination comprehensive medical therapy with ARNI commenced de novo on serum potassium. (a–c) Estimated treatment effects on serum potassium are shown for patients with HFrEF (a); all patients with HFmrEF/HFpEF (b) and patients with HFmrEF/HFpEF with LVEF < 60% (c). The center points and error bars represent the predicted treatment effects and 95% prediction intervals. The estimates for ARNI and its combinations are based on switching from existing ACEI/ARB therapy. All other treatments are compared to placebo. The mean predicted treatment effects are derived from individual drug class-specific regression equations, assuming additive effects of different drug classes after covariate adjustment. The prediction intervals are calculated with non-parametric

bootstrapping using the first-order normal approximation with 300 replicates. a–c apply the same drug class-specific regression models to the pooled average covariate profile of each population. The models were developed in a total of 27,330 participants with exact sample sizes for each drug class and the full model equations given in the Supplement. ARNI, angiotensin receptor blocker – neprilysin inhibitor; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; MRA, steroidal mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; PI, prediction interval; SBP, systolic blood pressure; SGLT2i, sodium glucose cotransporter 2 inhibitors.



Extended Data Fig. 6 | Estimated treatment effects of combination comprehensive medical therapy in HF on the composite endpoint of cardiovascular death and worsening HF events for de novo use of ARNI. (a–c) Estimated treatment effects on the composite endpoint of cardiovascular death and worsening HF events are shown for patients with HFrEF (a); all patients with HFmrEF/HFpEF (b) and patients with HFmrEF/HFpEF with LVEF < 60% (c). The center points and error bars represent the mean hazard ratios and 95% confidence intervals. All treatments are compared to placebo. The estimates for ARNI and its combinations are compared to a putative placebo. Treatment estimates are derived from 8,442 participants in PARADIGM-HF, 4,744

participants from DAPA-HF, 1,663 participants in RALES, 4,822 participants in PARAGON-HF; 6,263 participants in DELIVER; 1,767 participants from the Americas cohort of the TOPCAT trial, 6,001 participants from FINEARTS-HF, 2,028 participants from CHARM-Alternative and 3,023 participants from CHARM-Preserved. ARNI, angiotensin receptor blocker – neprilysin inhibitor; CI, confidence interval; HFrEF, heart failure with reduced ejection fraction; HFmrEF/HFpEF, heart failure with mildly reduced or preserved ejection fraction; sMRA, steroidal mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; SGLT2i, sodium glucose cotransporter 2 inhibitors.



Extended Data Fig. 7 | Short-term physiological effects and long-term clinical benefit of combination comprehensive medical therapy with ARNI commenced de novo. This figure compares the risk reductions in cardiovascular death and worsening HF with the short-term changes in SBP, eGFR and potassium using different combinations of medical therapy. Each point represents a treatment combination compared to placebo for the specified heart failure population. Center points and horizontal error bars represent the mean treatment effect and 95% prediction intervals for the predicted short-term change in blood pressure, eGFR and serum potassium. Center points and vertical error bars represent the mean treatment effect and 95% confidence intervals for the relative risk reduction in cardiovascular death or worsening heart failure. Colors denote number of drug classes: red = monotherapy, blue = dual therapy, green = triple therapy. Shapes distinguish individual drug combinations within

each therapy group as shown in the legend. Treatment estimates are derived from 8,442 participants in PARADIGM-HF, 4,744 participants from DAPA-HF, 1,663 participants in RALES, 4,822 participants in PARAGON-HF; 6,263 participants in DELIVER; 1,767 participants from the Americas cohort of the TOPCAT trial, 6,001 participants from FINEARTS-HF, 2,028 participants from CHARM-Alternative and 3,023 participants from CHARM-Preserved. ARNI, angiotensin receptor-neprilysin inhibitor; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFref, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction; nsMRA, non-steroidal mineralocorticoid receptor antagonist; SBP, systolic blood pressure; SGLT2i, sodium-glucose cotransporter-2 inhibitor; sMRA, steroidal mineralocorticoid receptor antagonist.

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give <i>P</i> values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No specific software were used for additional data collection for this pooled analysis of previously conducted randomised trials.
Data analysis	All analyses were performed with Stata version 18.1 and R version 4.1.2

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All trial funders are committed to sharing access to patient-level data and supporting clinical documents from eligible studies. Trial data availability is subject to the separate criteria and processes of AstraZeneca (<https://astrazenecagrouptrials.pharmacm.com/ST/Submission/Disclosure>), Bayer (<https://vivli.org/ourmember/bayer/>), Novartis (https://www.novartis.com/sites/novartis_com/files/clinical-trial-data-transparency.pdf) and Pfizer (<https://www.pfizer.com/science/clinical-trials/data-and-results>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Yes. We have provided sex specific reporting on outcomes including blood pressure, kidney function and serum potassium.
Reporting on race, ethnicity, or other socially relevant groupings	This study does not report any specific race, ethnicity or socially relevant groupings
Population characteristics	See Table 1
Recruitment	This is a pooled analysis of separate individual trials. The recruitment strategy for each trial are separately published
Ethics oversight	All the included trials were approved by ethics committees or institution review boards at all participating sites and all participants provided written informed consent. Ethics oversight was obtained for each individual trial and has been published separately.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	This was a pooled analysis of 38,753 participants (16,877 with HFrfEF and 21,876 with HFmrEF/HFpEF) across 9 randomized trials.
Data exclusions	Participants from TOPCAT who were enrolled in Russia and Georgia were excluded due to previously reported concerns that participants enrolled in Russia and Georgia did not have symptomatic HF and may not have been taking study medications.
Replication	We performed external validation of our model as an attempt to replicate our findings.
Randomization	This is a post-hoc analysis. Randomisation in each respective trial has been published previously and is outlined in the Methods.
Blinding	All included trials were double-blinded.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	PROSPERO CRD420261299819
Study protocol	Full protocol can be found in this manuscript.
Data collection	Full details of data collection for each trial can be found in each previous trial protocol.
Outcomes	Key measures evaluated were changes in SBP, DBP, eGFR, and serum potassium levels. Additionally, we assessed the composite endpoint of cardiovascular death and first worsening HF event to estimate the expected treatment benefits from combination therapy. The definition for worsening heart failure can be found in Extended Data Table 1.

Plants

Seed stocks	<i>Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.</i>
Novel plant genotypes	<i>Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.</i>
Authentication	<i>Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.</i>